Rohan Kumar

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Education

University of Illinois at Urbana-Champaign — **Expected Graduation Date: December 2026**
Bachelor of Science in Computer Science and Statistics (GPA: 3.86 / 4.00) — *Urbana-Champaign Area, IL*
Coursework: Object Oriented Programming, Data Structures, Algorithms, Database Systems, Computer Systems, Distributed Systems, High Frequency Trading Technology, Statistical Modeling, Statistical Learning

Skills

Programming Languages: Python, C#, C/C++, Java, TypeScript, JavaScript, SQL, R, HTML/CSS
Frameworks & Libraries: React, React Native, Node.js, .NET, Flask, Django, TensorFlow, Scikit-Learn, Pandas, Numpy
Cloud & DevOps: AWS (Lambda, EC2), Google Cloud Platform (GCP), Azure, Docker, Kubernetes, Git, GitHub Actions
Tools & Concepts: RESTful APIs, Agile, Test-Driven Development, LLMs, Vector Databases (Pinecone)

Experience

Software Engineer Intern — February 2026 – April 2026
Amazon (Incoming Winter 2026) — *Seattle, WA*

Software Engineering Research Assistant — January 2024 – December 2025
University of Illinois at Urbana-Champaign — *Urbana-Champaign Area, IL*
- Engineered a scalable data pipeline in Python to process and analyze teamwork behaviors across 100+ GitHub repositories.
- Architected a cloud-native backend on GCP and Firebase, ensuring efficient data management for over 120 concurrent users.
- Developed and deployed a responsive dashboard using Next.js and FastAPI with real-time data tracking capabilities.
- **Technologies:** Python, Numpy, Pandas, Sci-Kit Learn, Next.js, FastAPI, GCP, Firebase, RESTful APIs

Software Engineer Intern — May 2025 – August 2025
Relativity — *Chicago, IL*
- Owned the end-to-end development of internal API extensions and automated GitHub Actions CI/CD workflows, reducing direct client data access during incident resolution and improving resolution speed.
- Designed and engineered fault-tolerant .NET (C#) migration jobs to transition over 1TB of data from legacy SQL systems to a distributed, Azure-hosted NoSQL architecture, enhancing horizontal scalability and reducing storage costs.
- **Technologies:** C#, .NET, Azure Kubernetes Service (AKS), MySQL, Docker, RESTful APIs, GitHub Actions

Software Engineer Intern — June 2024 – July 2024
AM Best — *Oldwick, NJ*
- Developed a production .NET API in C# to serve financial records and credit ratings for over 300 insurance clients.
- Identified and resolved performance bottlenecks, implementing caching strategies that reduced query response times by 25%.
- **Technologies:** C#, MySQL, Blazor, ASP.NET, Azure Services, RESTful APIs, CI/CD

Projects

PAUL LLM Assistant – Illinois Business Consulting (IBC) | *Technical Lead*
- Designed, developed, and containerized an internal LLM assistant using Python and Docker to enhance workflow efficiency for 250+ active consultants by intelligently answering complex organizational queries.
- Engineered a prompt classification system to route queries to either the OpenAI API or a Pinecone vector database, enabling semantic search across past project data to generate context-aware, case-relevant responses.
- **Technologies:** Python, OpenAI API, Pinecone, Scikit-Learn, FastAPI, Docker, Large Language Models (LLMs)

Live Staffing System – Illinois Business Consulting (IBC) | *Technical Lead*
- Engineered a real-time drafting lobby using Socket.io web sockets in a full-stack application, streamlining the assignment of 250+ active consultants to client projects owned by different Senior Managers.
- Deployed the application on Google Cloud Platform, migrating from Cloud Run to Compute Engine for greater control and scalability, and managed all user and consultant data using CloudSQL (Postgres).
- **Technologies:** Python, FastAPI, JavaScript, Socket.io, GCP, Cloud Run, Compute Engine, CloudSQL, PostgreSQL

Market Data Feeds Compression (PCAP Compression) – FinTech Lab @ UIUC | *Software Engineer*
- Designed and implemented a high-performance C++17 market data system for IEX TOPS/DEEP and NASDAQ ITCH PCAP feeds, using Zstandard (Zstd) with dictionary training to reduce data size by 80-85% (≈15% gain over generic Zstd).
- Engineered a streaming, memory-efficient architecture (≤ 256MB) capable of processing multi-GB captures at 40K+ packets/sec, suitable for continuous market-hours ingestion while maintaining fast decompression and system reliability.
- **Technologies:** C++17, libpcap, Zstandard (Zstd), Make, Linux, Data Infrastructure, Systems Programming

Leadership

Technology Director — May 2025 – December 2025
National Organization for Business and Engineering (NOBE): Illinois Chapter — *Urbana-Champaign Area, IL*
- Led a team of 40+ student software engineers across 4+ technical projects for a diverse portfolio of clients.